WORKED SOLUTION

STEP I Specimen paper, Question 6

Pure • Euclidean geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q6

The key idea

The coin is a regular heptagon with a circular "cap" bulging out of each side. So its area is the heptagon plus seven circular segments, and a segment is a sector minus a triangle. Everything then comes down to two angles: the angle each arc subtends at its centre (the inscribed angle theorem gives it at once) and the angle between a long diagonal of the heptagon and the radius of its circumcircle.

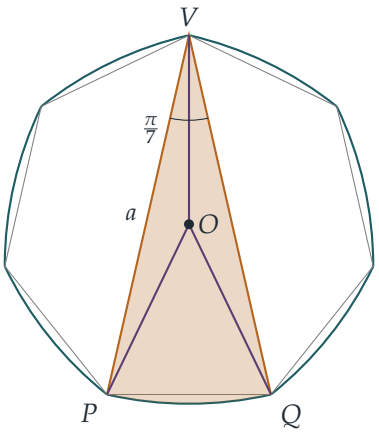
Solution

The shape

Each vertex is at distance a from the two ends of the arc opposite it (both ends lie on the circle of radius a centred there). So the seven "long" diagonals, each joining a vertex to one three places round, all have length a. Equal arcs of equal radius subtend equal angles at their centres, so the seven sides (the chords of the arcs) are also equal.

The long diagonals, taken in turn, form a seven-pointed star with all sides equal to a. Its angle at each point is the angle between the two diagonals from that vertex, and those diagonals, together with the side opposite, form a triangle with sides a, a and the common side length. All seven such triangles are congruent, so the star is equiangular as well as equilateral, hence regular. Therefore the seven vertices are the vertices of a regular heptagon, with centre O and circumradius R, say.

Label one vertex V, and let P and Q be the ends of the arc centred at V; P and Q are adjacent vertices of the heptagon.



The angle of each arc

The side PQ subtends an angle $2\pi/7$ at the centre O. Since V lies on the circumcircle, on the major arc PQ, the inscribed angle theorem gives

$$\angle PVQ = \tfrac{1}{2} \cdot \frac{2\pi}{7} = \frac{\pi}{7}.$$

So each arc is part of a circle of radius a subtending angle $\pi/7$ at its centre, and the sector VPQ (shaded) has area

$$\tfrac{1}{2}a^2 \cdot \frac{\pi}{7} = \frac{\pi a^2}{14}.$$

Adding up the area

The face of the coin consists of the heptagon together with seven segments, one on each side. The heptagon splits into seven triangles like OPQ, and the segment on PQ is the sector VPQ minus the triangle VPQ. Hence

$$\begin{aligned}\text{Area} &= 7\,[OPQ] + 7\big(\text{sector } VPQ - [VPQ]\big) \\ &= \frac{\pi a^2}{2} - 7\big([VPQ] - [OPQ]\big),\end{aligned}$$

where $[\,\cdot\,]$ denotes area.

Ask yourself How can we find $[VPQ] - [OPQ]$ without working out either triangle separately?

The centre O lies inside triangle VPQ, on its axis of symmetry, so removing triangle OPQ from triangle VPQ leaves the two congruent triangles OVP and OVQ:

$$[VPQ] - [OPQ] = [OVP] + [OVQ] = 2 \cdot \tfrac{1}{2}\, R\, a \sin \angle OVP.$$

By symmetry VO bisects $\angle PVQ$, so $\angle OVP = \pi/14$. Triangle OVP is isosceles with $OV = OP = R$, so its base angles are both $\pi/14$, and dropping the perpendicular from

O to VP gives

$$a = VP = 2R\cos\frac{\pi}{14}, \qquad \text{so} \qquad R = \frac{a}{2\cos(\pi/14)}.$$

Therefore

$$[VPQ] - [OPQ] = R\,a\sin\frac{\pi}{14} = \frac{a^2\sin(\pi/14)}{2\cos(\pi/14)} = \frac{a^2}{2}\tan\frac{\pi}{14},$$

and

$$\text{Area} = \frac{\pi a^2}{2} - \frac{7a^2}{2}\tan\frac{\pi}{14} = \frac{a^2}{2}\left(\pi - 7\tan\frac{\pi}{14}\right),$$

as required.

Checks

- Numerically, $\frac{1}{2}(\pi - 7\tan\frac{\pi}{14}) \approx 0.7719$, so the area is about $0.772a^2$. Sampling the seven arcs densely and using the shoelace formula agrees to six decimal places, and so does a Monte Carlo estimate (a point is in the coin exactly when it is within a of all seven vertices).
- The coin has width a in every direction, like a disc of diameter a, whose area is $\pi a^2/4 \approx 0.785a^2$. Of all shapes of constant width a the disc has the greatest area, so we expect a little less, and $0.772a^2 < 0.785a^2$ as it should be.
- The same argument for a Reuleaux n-gon (n odd) gives $\frac{a^2}{2}(\pi - n\tan\frac{\pi}{2n})$; as $n \to \infty$, $n\tan\frac{\pi}{2n} \to \frac{\pi}{2}$ and the area tends to $\pi a^2/4$, the disc. For $n = 3$ it gives $\frac{1}{2}(\pi - \sqrt{3})a^2$, the familiar Reuleaux triangle.

Common slips

- Taking the arc's angle to be $2\pi/7$ (the angle at the centre O) rather than $\pi/7$ (the angle at the opposite vertex, where the arc's centre actually is).
- Confusing a with the circumradius R: a is a long diagonal, $a = 2R\cos(\pi/14)$.

Beyond STEP

The 50p coin is a curve of *constant width*: every pair of parallel tangent lines is the same distance a apart, which is why a coin slot can measure it the same way however it is turned. Barbier's theorem says every such curve has perimeter πa (here $7 \cdot a \cdot \pi/7$), and the Blaschke–Lebesgue theorem says the Reuleaux triangle has the least area. For more on this kind of geometry, Courant and Robbins, *What Is Mathematics?*, has a lovely section on curves of constant width.

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